LISA CUNEO

PERSONAL INFORMATION

Place of Birth	Genova, Italy
Date of Birth	25 July 1995
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PERSONAL INTRODUCTION

My studies have focused on applied mathematics. My Bachelor and Master thesis concerned inverse problems, in particular the spatial resolution of Electro/Magneto-encephalography and, a widely used brain image techniques. I took part to several Schools on biomedical data analysis and Machine Learning. During the two stages at Axpo, I have increased my skills of data analysis, developing algorithms for anomaly detection via model design. I spent six months at the Department of Neuroscience and Biomedical Engineering (NBE) at Aalto University, Helsinki (Finland), within the Erasmus+ program, to work on my Master thesis. I taken the Phd in Physics and nanosciences - bio-nanosciences curriculum. During the period abroad in Cambridge (UK) I had the opportunity to solidify my understanding of common deep learning architectures and acquire new skills in the field of data analysis and generative models. Now I'm undertaking a Post Doc at the IIT between the Molecular Microscopy and Spectroscopy and the Computational Statistics and Machine Learning groups.

TECHNICAL EXPERTISE

- Biomedical data
- Data analysis
- Computer Science
- Inverse Problem
- Machine Learning
- Microscopy imaging

PROFESSIONAL EXPERIENCE

Jun - Aug 2019 — Axpo – Genova, Italy

Stage Axpo, Risk Management

Analysis of transactions to automatically check if their modification processes have been carried out in compliance with the policies. The stage was focused on pattern recognition, model design, anomaly detection.
Reference: Ing. Fabrizio Pischedda · fabrizio.pischedda@axpo.com

Jan - Jun 2019 — INSPIRE S.R.L – Genova, Italy

C.Co.C. INSPIRE S.R.L, Start Up

Applied scientific research that aims to create and test an innovative land monitoring system, through the use of drones. Creation of Matlab software for testing and demonstrating the identified algorithm. Search for the most efficient constrained scheduling solution for multiple requests.
Reference: Ing. Marco Ghio · m.ghio@be-inspire.com

Mar - May 2019 — Axpo – Genova, Italy

Internship Axpo, Risk Management

Validation of commodity prices coming from a model. Implementation of an algorithm to detect anomalies in historical series of forward market prices.
Reference: Ing. Fabrizio Pischedda · fabrizio.pischedda@axpo.com

EDUCATION

2024-On going Istituto Italiano di Tecnologia (IIT) – Genova, Italy

Post Doc in Microscopy and Machine Learning

I'm part of the *Molecular Microscopy and Spectroscopy* and *Computational Statistics and Machine Learning* groups of the Istituto Italiano di Tecnologia (IIT) under the supervision of Giuseppe Vicidomini and Massimiliano Pontil.

2020-2024 Università degli studi di Genova & IIT – Genova, Italy

PhD in Physics and Nanosciences

I was part of the *Nanoscopy* & NIC@IIT group of the Istituto Italiano di Tecnologia under the supervision of Prof. Alberto Diaspro.
Graduation date: 16/05/2024
My reasearch is focused on computational analysis of biomedical images.

May - Sep. 2023 University of Cambridge – Cambridge, United Kingdom

Visitor student

Visitor student at the Computational Biology at Department of Computer Science and Technology under the supervision of Prof. Pietro Liò.

Sep. 2019 - Feb. 2020 Aalto University – Helsinki, Finland

Erasmus+ Traineeship

Visitor student at the Neuroscience and Biomedical Engineering (NBE) department under the supervision of Prof. Lauri Parkkonen.

2019-2020 Università degli studi di Genova – Genova, Italy

PERCORSO FORMATIVO 24 CFU

Course to obtain the 24 credits in the anthropo-psycho-pedagogical disciplines and in the methodologies and educational technologies required for participation in the competition for public schools as per Ministerial Decree 257/2017.

2017-2020 Università degli studi di Genova – Genova, Italy

Masters of Applied Mathematics

Thesis Title: *An inverse problem perspective on the optimal design of sensor arrays for on-scalp MEG.*
Supervisors: Prof. Alberto Sorrentino and Prof. Lauri Parkkonen
Final degree mark: 110/110 cum laude
Graduation date: 25/03/2020

2014-2017 Università degli studi di Genova – Genova, Italy

Bachelor of Mathematics

Thesis Title: *Numeric study of spatial resolution of EEG through ordinary least squares problems.*
Supervisor: Prof. Alberto Sorrentino
Final degree mark: 102/110
Graduation date: 15/11/2017

TEACHING ASSISTANT

2021-2023 Università degli studi di Genova – Genova, Italy

Mathematical Analysis module 2

Exercise classes for "*Mathematical Analysis module 2*" course, cod. 60235 SSD MAT/05, I semester, year 2 of the Degree in Mechanical Engineering
Supervisor: Prof. Edoardo Mainini

2021-2023 Università degli studi di Genova – Genova, Italy

Statistics

Exercise classes for "*descriptive statistics*" course, cod. 52480 SECS - S/01, II semester, year 1 of the Degree in mathematics and SMID
Supervisor: Prof. Alberto Sorrentino and Francesco Porro

2022-2023 Università degli studi di Genova – Genova, Italy

Mathematical Analysis 2

Exercise classes for "*Mathematical Analysis* " course, cod. 25900 SSD MAT/05, I semester, year 2 of the Degree in Mathematics
Supervisor: Prof. Giovanni Alberti

2024-2026 Università degli studi di Genova – Genova, Italy

Applied Statistics 1 Exercise classes for "*applied statistics*" course, cod. 108582 SECS - S/01, II semester, year 2 of the Degree in mathematics and SMID (Statistica Matematica e Trattamento Informatico dei Dati)
Supervisor: Prof. Eva Riccomagno and Francesco Porro

REVIEW ACTIVITY

Reviewer for:

- Microscopy Research and Technique (MRT), Wiley

PUBLICATIONS

Published Medical Image Analysis
Hunting imaging biomarkers in pulmonary fibrosis: Benchmarks of the AIIB23 challenge.
Guang Yang, Michail Mamalakis, Lisa Cuneo, Francesco Prinzi, Gianluca Carlini et all
DOI: https://doi.org/10.1016/j.media.2024.103253.

Published Biophysical Journal
Remodeling of the nuclear landscapes and epigenome during retro-transformation of neuroblastoma cells mediated by the non-coding RNA NDM29
Francesca Baldini, Lama Zeaiter, Lisa Cuneo, Paolo Bianchini, Laura Vergani, Aldo Pagano, Alberto Diaspro
DOI: https://doi.org/10.1016/j.bpj.2023.11.561.

Published Biochimica et Biophysica Acta (BBA) - Molecular and Cell Biology of Lipids
Nuclear and chromatin rearrangement associate to epigenome and gene expression changes in a model of in vitro adipogenesis and hypertrophy.
Francesca Baldini, Lama Zeaiter, Farah Diab, Hawraa Zbeeb, Lisa Cuneo, Aldo Pagano, Piero Portincasa, Alberto Diaspro, Laura Vergani
DOI: https://doi.org/10.1016/j.bbalip.2023.159368.

Published IL NUOVO CIMENTO
An automated tool to estimate chromatin compaction in stained nuclei.
Lisa Cuneo, Marco Castello, Francesca Baldini, Alberto Diaspro
DOI: https://doi.org/10.1393/ncc/i2022-22193-5.

Published IL NUOVO CIMENTO
Investigating nanoscale chromatin alterations involved in neuroblastoma transformation by optical nanoscopy.
Francesca Baldini, Isotta Cainero, Lisa Cuneo, Michele Oneto, Elena Gatta, Chantal Usai, Aldo Pagano, Laura Vergani, Alberto Diaspro
DOI: https://doi.org/10.1393/ncc/i2022-22191-7.

Published IL NUOVO CIMENTO
A deep learning-based method to spectrally separate overlapping fluorophores based on their fluorescence lifetime.
Lisa Cuneo, Marco Castello, Simonluca Piazza, Irene Nepita, Isotta Cainero, Giorgio Tortarolo, Luca Lanzanò, Paolo Bianchini, Giuseppe Vicidomini, Alberto Diaspro
DOI: https://doi.org/10.1393/ncc/i2023-23130-x.

Published La rivista del nuovo cimento
Emerging Mueller matrix microscopy applications in biophysics and biomedicine.
Alberto Diaspro, Paolo Bianchini, Fabio Callegari, Lisa Cuneo, Riccardo Marongiu, Aymeric Le Gratiet, Ali Mohebi, Marco Scotto, Colin J. R. Sheppard
DOI: https://doi.org/10.1007/s40766-023-00046-5.

Conference proceedings EPJ Web Conf. Volume 287, EOS Annual Meeting (EOSAM), September 2023
The Artificial Microscope.
Alberto Diaspro, Paolo Bianchini, Lisa Cuneo
DOI: https://doi.org/10.1051/epjconf/202328713012.

Conference proceedings The Italian Physical Society - Proceedings of the International School of Physics "Enrico Fermi" - Volume 210: Multimodal and Nanoscale Optical Microscopy - pp. 143 – 145 - September 2023
Scattering Networks: a tool to remove the background.
Lisa Cuneo, Simone Civita, Paolo Bianchini, Alberto Diaspro
DOI: https://doi.org/110.3254/ENFI230014.

Conference proceedings Biophysical Journal 122 (3) pp. 462a February 2023
A deep learning method to separate fluorophores based on their fluorescence lifetime.
Lisa Cuneo, Marco Castello, Simonluca Piazza, Irene Nepita, Luca Lanzanò, Paolo Bianchini, Giuseppe Vicidomini, Alberto Diaspro
DOI: https://doi.org/10.1016/j.bpj.2022.11.2483.

Conference proceedings Biophysical Journal 121 (3) pp. 476a February 2022
Decrypting nanoscale chromatin architecture alterations implicated in neuroblastoma transformation by optical nanoscopy.
Francesca Baldini, Isotta Cainero, Lisa Cuneo, Irene Nepita, Chantal Usai, Paolo Bianchini, Laura Vergani, Aldo Pagano, Alberto Diaspro
DOI: https://doi.org/10.1016/j.bpj.2021.11.402.

Conference proceedings Biophysical Journal 121 (3) pp. 362a February 2022
Investigating the role of chromatin compaction at the nanoscale in Hutchinson-Gilford progeria syndrome using expansion microscopy
Chantal Usai, Isotta Cainero, Lisa Cuneo, Francesca Baldini, Matteo Mariangeli, Irene Nepita, Paolo Bianchini, Alberto Diaspro
DOI: https://doi.org/10.1016/j.bpj.2021.11.938.

Conference proceedings Technical Digest Series (Optica Publishing Group), paper JTh5A.111 November 2021
An Automated Tool to Analyse 3D Fluorescence Images of Stained Nuclei.
Lisa Cuneo, Francesca Baldini, Marco Castello, Irene Nepita, Simonluca Piazza, Laura Vergani, Alberto Diaspro
DOI: https://doi.org/10.1364/FIO.2021.JTh5A.111.

Under submission Medical Image Analysis
Regularizations for ISM inverse problem
Sofia Agostoni, Lisa Cuneo, Christian Daniele, Giacomo Garrè, Alessandro Zunino, Luca Calatroni, Giuseppe Vicidomini

In preparation Nature Methods
Independent noise realizations enable morphology-agnostic regularization for image deconvolution microscopy
Lisa Cuneo, Alessandro Zunino, Laurent Le, Sofia Agostoni, Saverio Salzo, Massimiliano Pontil, Luca Calatroni, Giuseppe Vicidomini

In preparation SIAM Imaging
Regularization by Independent Noise Realizations: A Variational Model and a Convergent Relaxed Block Proximal-Gradient Method
Lisa Cuneo, Saverio Salzo, Giuseppe Vicidomini, , Luca Calatroni, Massimiliano Pontil

Under review Scientific Reports
Advanced Background Removal Methods in Single Molecule Localization Microscopy Using Scattering Networks and SVD
Lisa Cuneo, Simone Civita, Ivan Trapasso, Luca Ratti, Paolo Bianchini, Alberto Diaspro

AWARDS

Challenge MICCAI AIIB23-Task 1 Segmentation: Airway-Informed Quantitative CT Imaging Biomarker for Fibrotic Lung Disease. Position: 8/20.
Link: Task1

Challenge MICCAI AIIB23-Task 2 Binary classification: Airway-Informed Quantitative CT Imaging

Biomarker for Fibrotic Lung Disease. Position: 4/6.
Link: Task2

Travel Grant SIBPA (Società Italiana di Biofisica Pura e Applicata) grant for 1500€ to participate at the BPS (BioPhysical Society) annual meeting at San Diego, California, USA.

Best communication The comunication with title ”An automated tool to estimate chromatin compaction in stained nuclei” was selected by the Scientific Committee for the Best Communications of 107° Congresso SIF.

Best communication The comunication with title ”Scattering Networks: a tool to remove the background” was selected by the Scientific Committee for the Best Flash talk of International School of Physics ”Enrico Fermi” summer school ”Multimodal and Nanoscale Optical Microscopy”.

Best communication The comunication with title ”A deep learning-based method to spectrally separate overlapping fluorophores based on their fluorescence lifetime.” was selected by the Scientific Committee for the Best Communications of 108° Congresso SIF.

INVITED PRESENTATIONS

March 7th, 2024 — PoliTO – Torino, Italy

Talk Big Data and Data Science Laboratory, DISMA, SmartData@PoliTO
Organized by: Ivan S. Trapasso.
Title: Scattering Networks and Singular Values Decomposition: different methods to remove background in Single-Molecule Localization Microscopy images.

July 21th, 2023 — CL – University of Cambridge, United Kingdom

Talk Artificial Intelligence Research Group Talks at the Computer Laboratory
Organized by: Prof. Pietro Liò.
Title: Scattering Networks and Singular Values Decomposition: different methods to remove background in Single-Molecule Localization Microscopy image.

June 5-6, 2023 — Stem Cell Institute – Cambridge, United Kingdom

Talk Cambridge Stem Cell Institute Annual Retreat, Industry Flash Talk.
Presentation of Genoa Instruments PRISM product.

May 23th, 2023 — CEB – University of Cambridge, United Kingdom

Talk Laser Analytics Group at the Department of Chemical Engineering and Biotechnology (CEB), Physics Troublemaking Meeting
Organized by: Prof. Clemens Kaminski.
Title: Scattering Networks and Singular Values Decomposition: different methods to remove background in Single-Molecule Localization Microscopy image.

Oct 12th, 2020 — Inverse Problems Seminar – Online

Seminar Inverse Problems Seminars.
Organized by: Inverse Problem group of Department of Mathematics and Statistics - University of Helsinki.
Presentation of my Master Thesis, whose topic was the optimization of an hypothetical on-scalp Magnetoencephalograph sensors configuration in terms of sensitivity and source localization.

COMPUTER SKILLS

Advanced: Matlab, *Python*, R, LaTeX
Intermediate: *Office*, *SQL*
Basic: *C++*, *Java*

LANGUAGE SKILLS

Mothertongue: Italian
B2: English

SOFT SKILLS

- *Abstract Thinking*
- *Problem Solving*
- *Lateral Thinking*
- *Teamwork*

SCHOOLS AND WORKSHOPS

CIL School 2026 *23 - 27 Feb, 2026 – Genova, Italy*
9th NIC@IIT Advanced Microscopy practical workshop.
Personal contribution: I collaborated in the organization of the event.
Website: NIC@IIT2024.

Nikon Workshop 2024 *2 - 6 Dec, 2024 – Genova, Italy*
Computational Imaging & Learning: From physics to data-driven methods.
Personal contribution: I collaborated in the organization of the event.
Website: CIL2026.

MALGA school *Sep. 2-6, 2024 – Genova, Italy*
Applied Harmonic Analysis and Machine Learning school, organized by: Machine Learning Genoa Center (MALGA)- Università degli Studi di Genova.
Website: ahaml2024.

Nikon Whorkshop 2023 *Nov. 27 - 1 Dec, 2023 – Genova, Italy*
8th NIC@IIT Advanced Microscopy practical workshop.
Personal contribution: I collaborated in the organization of the event.
Website: NIC@IIT2023.

Nikon Whorkshop 2022 *Nov. 28 - 2 Dec, 2022 – Genova, Italy*
7th NIC@IIT Advanced Microscopy practical workshop.
Personal contribution: I collaborated in the organization of the event.
Website: NIC@IIT2022.

MALGA school *Sep. 5-9, 2022 – Genova, Italy*
Applied Harmonic Analysis and Machine Learning school, organized by: Machine Learning Genoa Center (MALGA)- Università degli Studi di Genova.
Website: ahaml2022.

PRIMO Workshop 2022 *Sep. 6 - 8, 2022 – Trieste, Italy*
Post-graduate Researchers in Inverse problems, Machine learning, and Optimization.
I presented a talk with title ”Scattering Networks: a tool to remove background in microscopy images”.
Website: PRIMO.

School of Physics ”Enrico Fermi” *Jul. 10-15, 2022 – Varenna, Italy*
Multimodal and Nanoscale Optical Microscopy.
I presented a flash talk with title ”Scattering Networks: a tool to remove the background”.
Website: Varenna.

UMI-MIVA Winter School 2022 *Jan. - Feb., 2022 – Online*
Advanced methods for mathematical image analysis.
Website: MIVA.

Nikon Workshop *Nov. 30 - Dec. 3, 2021 – Genova, Italy*
6th NIC@IIT pratical workshop on advanced microscopy.
Website: NIC@IIT2021.

VaMOS Workshop *Nov. 22 - 23, 2021 – Online*
Advanced optimization methods for inverse problems & applications to image microscopy.
Website: VaMOS.

PRIMO Workshop 2021 *Oct. 11 - 13, 2021 – Online*
Post-graduate Researchers in Inverse problems, Machine learning, and Optimization.
Website: PRIMO.

NonInvasive Mathematics Workshop *Apr. 13 - 16, 2021 – Online*

Workshop bridging between mathematicians, engineers, neuroscientists, and clinicians working with magneto/electro-encephalography and similar techniques.
Website: DIMA.

"Convex Optimization" School *Jul. 20 - 24, 2020 – Online*

Organized by: Machine Learning Genoa Center - Università degli Studi di Genova.
Description: 20 hours advanced convex optimization course including theory classes and practical laboratory sessions.

"Deep Learning: A Hands-on Introduction" School *Jul. 13 - 17, 2020 – Online*

Organized by: Machine Learning Genoa Center - Università degli Studi di Genova.
Description: 20 hours advanced deep learning course including theory classes and practical laboratory sessions.

School of Neuroengineering *Jun. 18-22, 2018 – Genova, Italy*

7th Summer School of Neuroengineering 'Massimo Grattarola'.
Website: Neuroengineering.

Workshop "Invasive mathematics" *May. 11th, 2018 – Genova, Italy*

Workshop bridging between mathematicians, engineers, neuroscientists, and clinicians working with invasive brain imaging such as StereoEEG.

OTHER ACTIVITY

Projects During my studies I carry out the following projects:

- *Convex Optimization on Riemannian Manifolds*
  Final project for *Convex Optimization* course, 2021
- *Optogenetics*
  Final project for *Biophysics* course, 2021
- *Clustering of ElectroCorticoGraphy (ECoG) data during a visual categorization task*
  Final project for *Machine Learning* course, 2019
- *Curvatures and their properties*
  Final project for *Discrete Surface Analysis Methods and their applications* course, 2019
- *Numerical resolution of an elliptical PDE on bounded domain*
  Final project for *Numerical Treatment of Differential Equations* course, 2018

August 16, 2026

SIGNATURE

Lisa [illegible]